TABLE OF CONTENTS

ABSTRACT_

This project presents a cost-effective overheat detection system with automatic cutoff functionality, designed using the readily available LM358 dual operational amplifier (op-amp). The system utilizes a temperature sensor (LM35) to convert temperature changes into electrical signals. An op-amp within the LM358 acts as a comparator, continuously comparing the temperature sensor's signal with a pre-set reference voltage. When the surrounding temperature surpasses the designated threshold, the op-amp triggers a relay, initiating an automatic shut-off mechanism to safeguard the connected device. This report details the circuit design rationale, component selection process, operational principles, and potential applications of this system. The applications extend beyond personal computers and kitchens, encompassing various electronic devices like amplifiers and cooling systems for temperature-sensitive products. The overheat detector serves as a crucial safety measure by preventing overheating and potential damage to equipment or burning of contents.

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

The increasing reliance on electronic devices in both domestic and industrial settings has raised concerns about the risks associated with overheating. Overheating not only compromises the performance and longevity of electronic equipment but also possess significant safety hazards, including fire risks and potential damage to surrounding materials. Traditional overheat detection systems often come with high costs and complexity, limiting their accessibility, particularly in resource-constrained environments. Therefore, there is a growing need for cost-effective and efficient solutions to monitor and mitigate overheating issues in electronic devices.

1.2 OBJECTIVES

The primary objective of this project is to design and implement a cost-effective overheat detection system with automatic cutoff functionality. By utilizing readily available components such as the LM358 dual operational amplifier and the LM35 temperature sensor (or an NTC thermistor), the aim is to develop a system that can accurately detect temperature changes and trigger an automatic shutdown mechanism when overheating is detected. The project seeks to achieve a balance between affordability, simplicity, and reliability, making it accessible for various applications and environments.

1.3 RELEVANCE

This project is highly relevant due to the widespread use of electronic devices in both domestic and industrial settings. Overheating poses a significant risk to equipment performance, safety, and longevity, necessitating effective monitoring and mitigation measures. By developing a cost-effective overheat detection system, this project addresses critical safety concerns while also promoting the efficient use of resources. Additionally, the system's versatility enables its application in various contexts, ranging from personal computers and kitchen appliances to amplifiers and cooling systems for temperature-sensitive products.

1.4 REPORT ORGANIZATION

1. **Introduction:** Provides an overview of the project, its significance, and the objectives.

2. **Background**: Discusses the context and rationale behind the need for an overheat detection system, highlighting the risks associated with overheating in electronic devices.

3. **Component Selection:** Details the selection process for key components, including the LM358 dual operational amplifier and the LM35 temperature sensor (or an NTC thermistor), emphasizing their suitability for the project's objectives.

4. **Circuit Design:** Describes the design rationale and principles behind the overheat detection system, including the circuit diagram and operational principles.

5. **Implementation**: Provides a step-by-step guide on how to implement the overheat detection system, including component assembly and testing procedures.

6. **Results**: Presents the results of system testing, including performance metrics such as accuracy, response time, and reliability.

7. **Applications and Advantages**: Explores the diverse applications and advantages of the overheat detection system across different industries and environments.

9. **Conclusion**: Summarizes the project's findings, reiterates its significance, and suggests future research directions.

10. **References**: Lists all sources cited in the report.

CHAPTER 2

REVIEW OF LITERATURE

In the research paper[1], the researchers conducted experiments aboard manned spacecraft to explore the effects of microgravity on living organisms, particularly focusing on muscle atrophy, bone density loss, and cellular function alterations. They also investigated plant growth and development, finding significant impacts on root growth, nutrient uptake, and photosynthesis, which are crucial for long-term space missions. The controlled environment of spacecraft allowed for the study of microbial behaviour, including changes in virulence and antibiotic resistance. Despite challenges like limited resources and complex data interpretation, the findings have valuable applications for human health, agriculture, and future space exploration.

In paper[2], the authors explore various methodologies for diagnosing faults in overheat detection systems. The paper emphasizes the importance of accurate fault isolation and the role of sensing elements in detecting system malfunctions. The authors describe techniques for calculating short and open circuits and offer assistance in locating defective elements within the system. Their research highlights the necessity of robust diagnostic procedures to ensure the reliability and safety of overheat detection systems in industrial applications.

In paper[3], In recent years, overheating of equipment has become a significant issue in large enterprises, particularly within the petrochemical industry. The malfunctioning or partial breakdown of electrical, pneumatic, hydraulic, or electromechanical machinery often leads to a rise in operating temperatures, which can cause catastrophic failures. Traditional methods of monitoring temperatures lacked the precision and immediacy required to prevent such failures. Modern embedded systems, incorporating sensors like the LM35 temperature sensor and microcontrollers such as the ATMEGA328, offer real-time monitoring and alert capabilities. These systems measure temperature and provide alerts through various signals, including LEDs, buzzers, and SMS notifications via GSM modules, thereby enhancing safety and preventing damage by enabling prompt responses to overheating.

In paper[4], The literature on machine overheat detection systems highlights significant developments and advancements in this field. For instance, Tiwari and Shrivastava (2013) emphasize the necessity of such systems in industrial settings to reduce human labor and enhance efficiency by automating the monitoring process. Their work discusses an automated heat detector system that reduces manual monitoring and uses a program to calculate the number of free slots in machines based on switch presses. Similarly, Choubey and Sharma (2015) describe a temperature controller design that utilizes a Pt resistance thermometer for precise temperature measurement and a transistor preamplifier to control system operations. Recent advancements include the use of intelligent temperature control systems based on microcontrollers and digital temperature sensors to achieve high accuracy and precision in monitoring and controlling machine temperatures.

In paper[5], It provides an in-depth analysis of the application and characteristics of thermistors in automotive systems. It outlines the historical development of temperature sensors, highlighting the importance of accurate temperature measurement in enhancing automotive performance and emissions control. The paper focuses on PTC (Positive Temperature Coefficient) and NTC (Negative Temperature Coefficient) thermistors, explaining their nonlinear resistance-temperature relationship and the materials used in their construction. Various applications of these thermistors in engine cooling systems, intake air, transmission oil, and catalytic converters are discussed, emphasizing their role in engine management and HVAC systems. The experimental setup for measuring thermistor resistance and temperature is detailed, with results indicating the sensitivity and nonlinear behavior of these sensors. The paper concludes with discussions on recent advancements and the need for improved sensor accuracy using mathematical modeling and microprocessors.

CHAPTER 3

DESIGN CONSIDERATIONS

Overheat Detector with Auto Cut-Off System using Op-Amp (LM358)

Brief Introduction: This project presents a straightforward overheat detector with an automatic cut-off system utilizing a dual Op-Amp IC (LM358). The circuit functions as a heat sensor, activating an alarm (LED) when the ambient temperature surpasses a predetermined threshold. Additionally, it can completely deactivate the connected system upon overheating.

Components Required:

• Op-Amp	LM358	2	U2
• NPN Transistor	BC549	1	Q1
• Resistor	33kΩ	1	R1
• Resistor	1KΩ	4	R2,R3,R4,R5
• Temperature Sensor	LM35	1	U1
• SPDT Relay	12V	1	RL1
• LED		3	D5,D3,D2
• Capacitor	100uF	1	C1
• Power Supply	9V	1	Vcc

Pin Diagrams:

- LM358: Dual Op-Amp IC [U2]

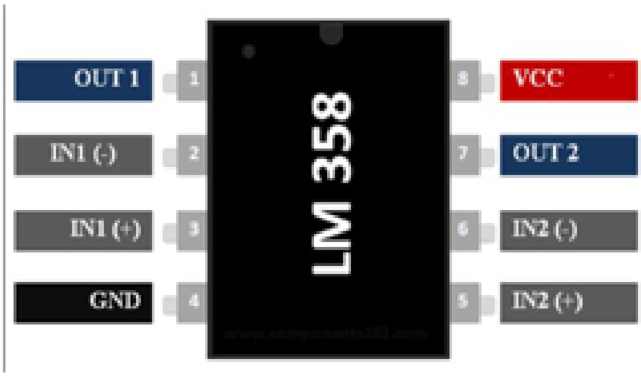

Fig-3.1 Dual Op-Amp LM358 IC

- Pin 1: Output A
- Pin 2: Inverting Input A
- Pin 3: Non-Inverting Input A
- Pin 4: Groud
- Pin 5: Non-Inverting Input B
- Pin 6: Inverting Input B
- Pin 7: Output B
- Pin 8: Vcc/V+

- Potentiometer: 1kΩ [

Fig-3.2 Potentiometer

- Pin 1: Vcc
- Pin 2: Output
- Pin 3: Ground

- Temperature sensor: LM35 [U1]

Fig-3.3 Temperature Sensor

- Pin 1: Vcc
- Pin 2: Output
- Pin 3: Ground

- PNP Transistor: BC549 [Q1]

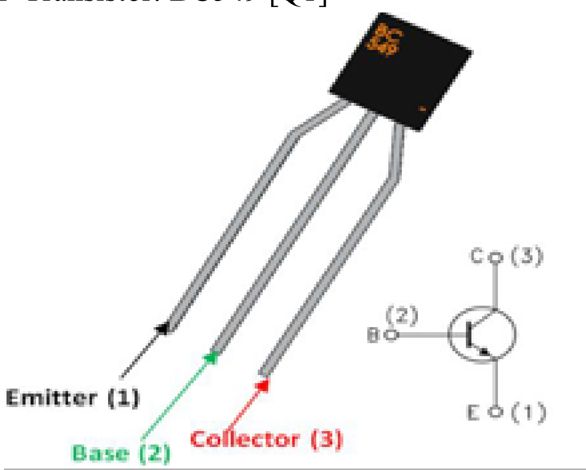

Fig-3.4 PNP Transistor

- Pin 1: Emitter
- Pin 2: Base
- Pin 3: Collector

- SPDT Relay: 12V Sugar cube Relay [RL1]

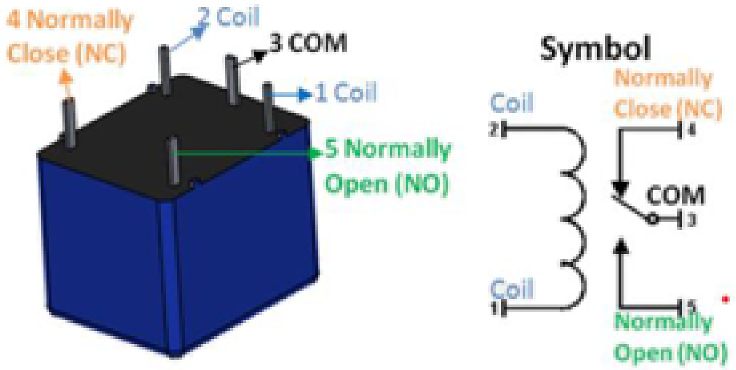

Fig-3.5 SPDT Relay

- Pin 1: Coil 1
- Pin 2: Common
- Pin 3: Coil 2
- Pin 4: NC
- Pin 5: NO

Mathematical Calculations:

- **LM35 Temperature Sensor (U1):**

  The LM35 temperature sensor outputs a voltage proportional to the temperature with a scale factor of 10 mV/°C. Given a temperature of 35°C:

  $V_{OUT} = 35 \times 10mV = 350mV = 0.35V$

- **Voltage Divider (R2 and R3):**

  The voltage divider does not act as a complete voltage divider because the capacitor here is being used to reduce the noise of the input current.

  $VR1 = Vin - VLM35 = 9V - 0.35V = 8.65V$

  $Vout = Vin - VR1 = 9V - 8.65V = 0.35V$

  $V_{OUT} = 0.35V$

- **Op-Amp Comparator (U2):**

<u>Voltage Divider (R2 and R3)</u>: The preset potentiometer sets the reference voltage (Vout) at the inverting input (pin 6) of the op-amp. The non-inverting input (pin 5) receives the output from the LM35 through a voltage divider

<u>Comparator Output Logic:</u>

The op-amp (U2) is configured as a comparator. The output of the comparator depends on the comparison between the voltages at its non-inverting (pin 5) and inverting (pin 6) inputs.

$V_{Threshold} = V_{out}$

1.When $V+ > V_{Threshold}$, the op-amp output will be high.

2.When $V+ \leq V_{Threshold}$, the op-amp output will be low.

In this case:

1. $V+ = 0.35V$
2. $V_{Threshold} = 0.175\ V$

Since 0.35 V > 0.175 V, the op-amp output goes high.

- The resistor values of 1kΩ and 33kΩ were chosen for specific roles in the Over Heat Detector with Auto Cut-Off System. R1 (1kΩ) filters noise and stabilizes the input voltage to the LM358 op-amp. R2 (1kΩ) and R3 (33kΩ) form a voltage divider setting the threshold voltage for the LM358 comparator at 0.32V, ensuring accurate detection of the desired temperature threshold. R2 (1kΩ) provides a standard reference voltage, while R3 (33kΩ) ensures the comparator activates reliably at lower temperatures, maintaining the circuit's operational integrity.

- **Transistor Base Current Calculation (Q1):**

When the op-amp output is high, it drives the base of the transistor Q1 through resistor R3.

$I_B = (V_{OUT} - V_{BE}) / R3$

The base-emitter voltage (V_{BE}) is typically considered to be around 0.7V for silicon transistors like the BC549, as this is a common value for the voltage drop across the base-emitter junction when the transistor is in its active region.

$I_B = (12V-0.7V)/1k\Omega$

$I_B = 11.3mA$

- **Collector Current Calculation (Q1):**

  Using the current gain (β) of the transistor (assumed to be 100):

  $I_C = \beta \times I_B$

  $I_C = 100 \times 11.3mA$

  $I_C = 1130\ mA = 1.13A$

Reason for Using β=100 for BC549 Transistor:

1. **Transistor Characteristics**:
   - The BC549 transistor is a general-purpose NPN transistor commonly used in low-power amplification and switching applications.
   - Its typical current gain (β) ranges from 110 to 800 depending on the collector current and other operating conditions.
2. **Common Use Case**:
   - In many circuits, especially those involving switching applications like driving a relay, a conservative estimate of β=100 is often used to ensure reliable operation across a range of conditions.
3. **Simplification and Reliability**:
   - Using β=100 simplifies calculations and provides a margin of safety, ensuring that the base current calculation is sufficient to drive the transistor into saturation (fully on state).

- It ensures that there is enough current gain to activate the relay reliably.

- **Relay Activation:**

  The collector current I_C flows through the relay coil, activating it. The relay is rated for 12V, so it will operate correctly when I_C is sufficient to energize the coil.

  - The relay coil current required for operation is typically in the range of 100-200mA.
  - With I_C =1.13A, Q1 can provide sufficient current to energize the relay coil.

CHAPTER 4

CIRCUIT DIAGRAM

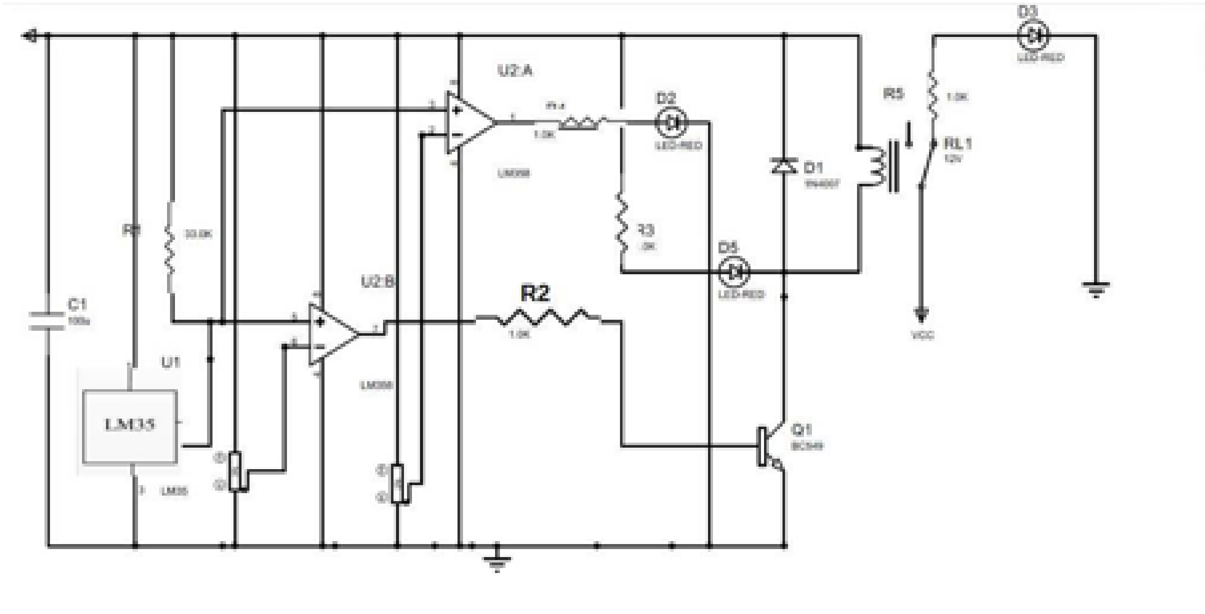

Fig - 4.1 Schematic Diagram

Working:

Temperature Sensing:

The LM35 (U1) senses the temperature and provides an analog voltage output proportional to the temperature (10 mV per degree Celsius).

In the circuit, the voltage output of LM35 is connected to the non-inverting input (pin 5) of the operational amplifier U2

Reference Voltage Setting:

A reference voltage is set using a preset potentiometer. This reference voltage is fed to the inverting input (pin 6) of the operational amplifier U2.

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Comparing Temperatures:

The operational amplifier U2 compares the voltage from the LM35 with the reference voltage. If the temperature exceeds the set threshold (the reference voltage), the output of U2 (pin 7) goes high, turning on the NPN transistor Q1.

Activating the Relay:

When Q1 is turned on, it energizes the relay RL1, causing it to switch. This can be used to cut off the power to a connected device, preventing it from overheating.

The diode D1 is used to protect the transistor Q1 from the back EMF generated when the relay coil is de-energized.

Indicators:

LED D2 is connected to the output of U2, which is configured as a comparator with a fixed reference. It turns on to indicate when the temperature exceeds the threshold. LED D5 is connected in parallel with the relay coil and turns on to indicate that the relay is activated. LED D3, connected to the normally closed contact of the relay, indicates the normal operation (i.e., when the relay is not activated).

Power Supply:

The circuit is powered by a DC supply, and the relay is designed for a 12V operation.

CHAPTER 5

IMPLEMENTATION AND RESULTS

1. **Software Implementation**

- D3 is on when the temperature is below **35°C**:

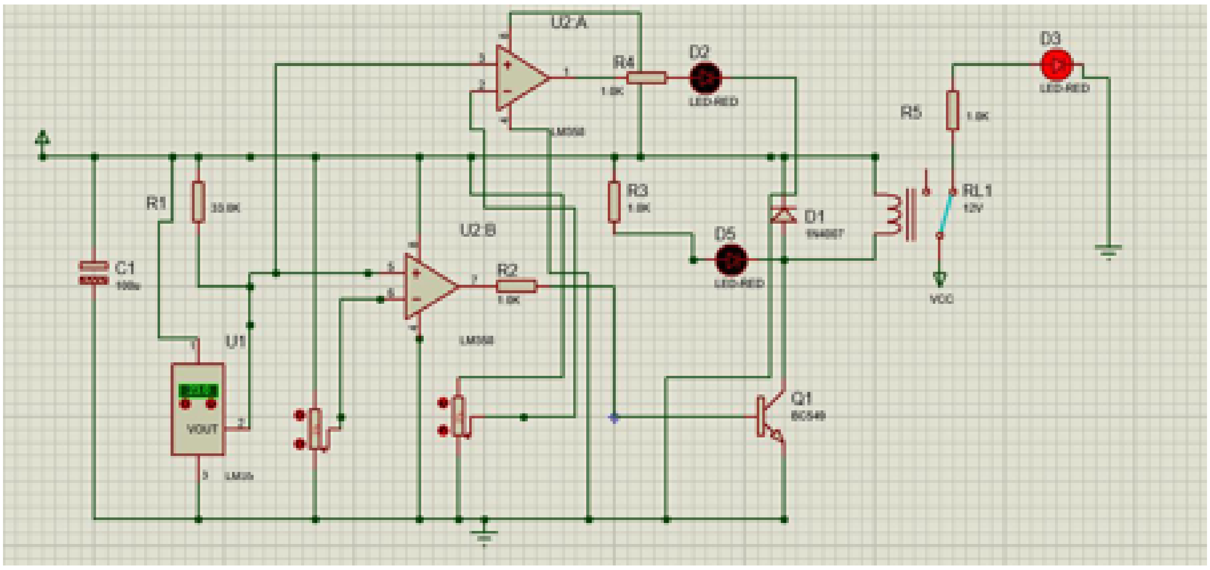

Fig - 5.1.1 Temperature less than **24°C**

- D2 turns on when temperature crosses **24°C**, this is an warning light

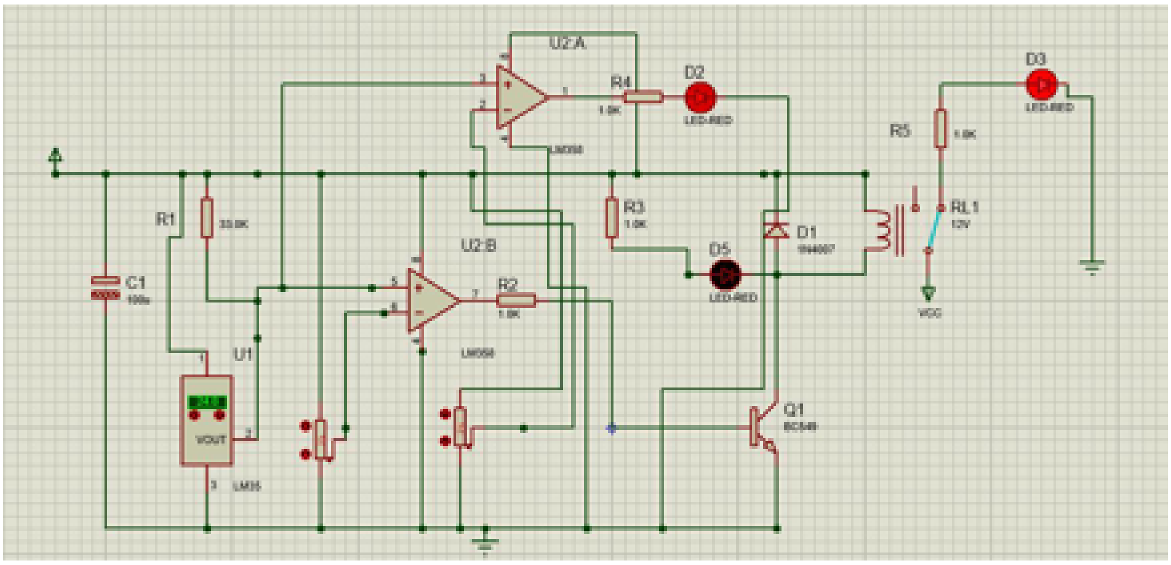

Fig 5.1.2 Temperature between **24°C** and **35°C**

- D3 gets turned off(cutting off the supply) and D5 becomes on as the temperature is now above **35°C** i.e, **36°C**.

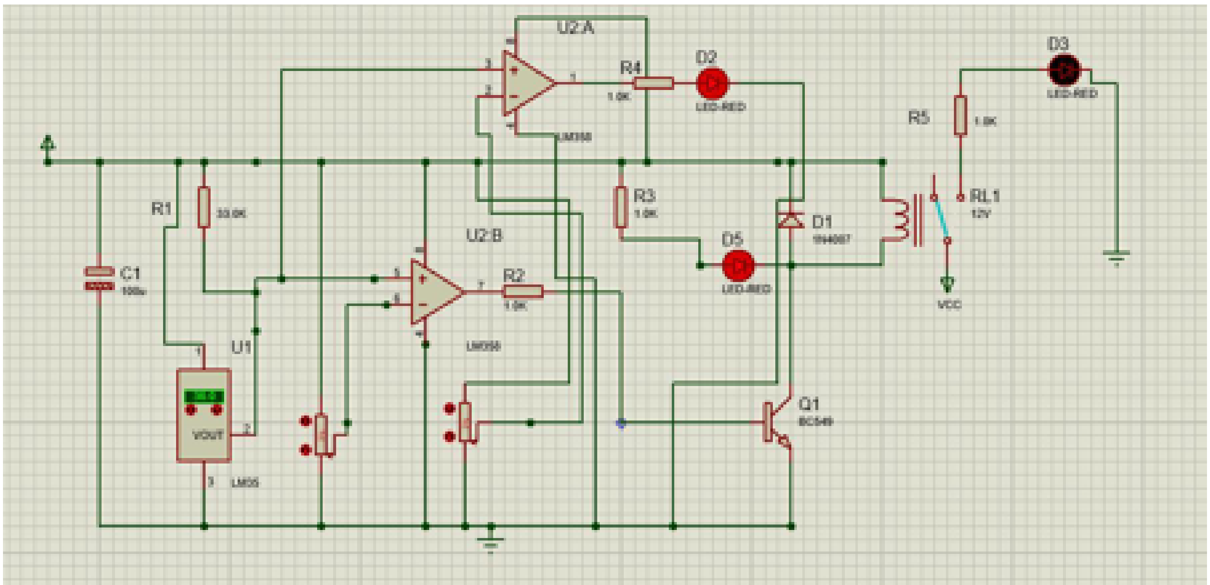

Fig - 5.1.3 Temperature greater than **35°C**

2. Hardware Implementation

- Initial stage when temperature crosses the **35°C**

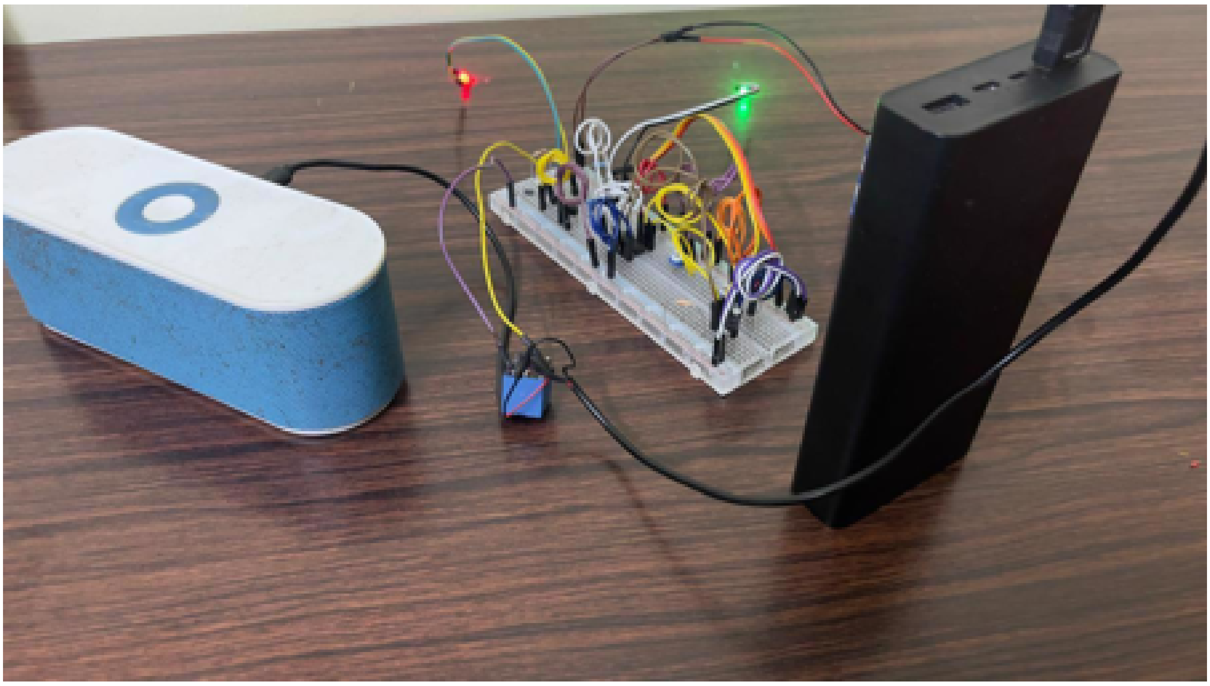

Fig 5.2.1 Initial stage when temperature crosses the **35°C**

- When the temperature is below **35°C**

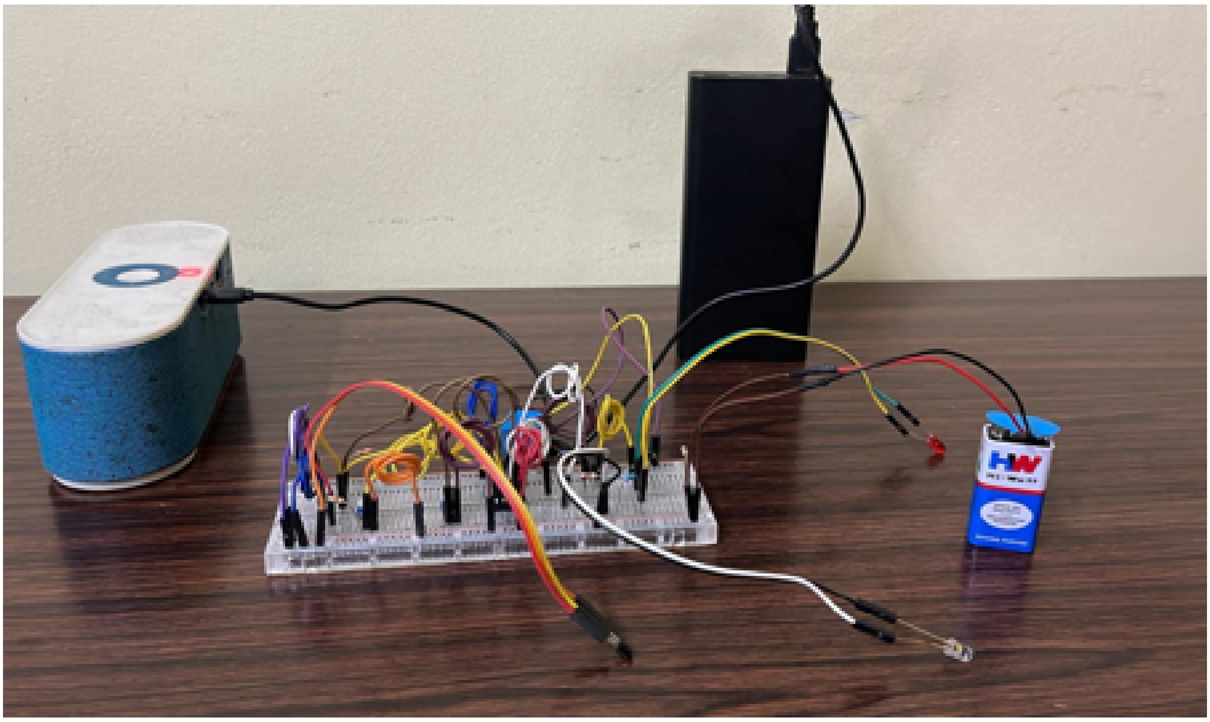

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*Fig - 5.2.2 When the temperature is below **35°C***

CHAPTER 6

ADVANTAGES & APPLICATIONS

Advantages:

1. **Automatic Cut-off**: The system can automatically turn off the heat source, preventing potential hazards and energy wastage.

2. **High Sensitivity**: Utilizes an LM358 operational amplifier for precise temperature detection and response.

3. **Cost-effective**: The components used in the circuit are inexpensive and readily available.

4. **Compact Design:** The circuit can be implemented on a small PCB, making it suitable for integration into various devices.

5. **Energy Efficient:** The system only activates the relay when necessary, conserving power.

6. **Reliability:** Components like the BC549 transistor and 1N4007 diode are known for their reliability and long lifespan.

Applications:

1. **Home Appliances**: Can be used in devices like ovens, heaters, and toasters to prevent overheating.

2. **Industrial Equipment**: Suitable for machinery that operates at high temperatures, providing safety and efficiency.

3. **Automotive**: Can be integrated into car engines to monitor and control engine temperature.

4. **HVAC Systems**: Helps in maintaining the desired temperature by controlling heating elements in HVAC systems.

5. **Fire Safety Systems**: Acts as a part of fire alarm systems to detect and respond to abnormal heat levels.

6. **Electronics**: Protects sensitive components in electronic devices from overheating by controlling cooling systems or shutting down the device.

CHAPTER 7

CONCLUSION & FUTURE SCOPE

Conclusion:

The Overheat Detector with Auto Cut-Off System is a vital safety feature designed to provide a cost-effective, reliable, and efficient solution for temperature monitoring and control. Utilizing easily accessible components such as the LM358 operational amplifier and the LM35 temperature sensor, the system ensures precise temperature detection and response, thereby preventing overheating and potential fire hazards. The automatic cut-off mechanism significantly enhances safety by automatically disconnecting the power supply to the heat source once the temperature exceeds a preset threshold. This project demonstrates the successful implementation of a compact and energy-efficient overheat detection system, which can be integrated into a variety of electronic devices and industrial applications, ensuring both equipment safety and longevity.

Future Directions:

Future advancements in this project could focus on several areas to enhance its functionality and applicability:

1. **Integration with IoT:** Incorporate Internet of Things (IoT) capabilities to enable remote monitoring and control, allowing users to receive real-time temperature alerts and manage the system through a mobile app or web interface.
2. **Enhanced Accuracy:** Implement more advanced sensors and microcontrollers to improve the accuracy and response time of the temperature detection system. This could include digital sensors with higher precision and faster processing speeds.
3. **Adaptive Thresholds:** Develop an adaptive threshold mechanism that adjusts the cutoff temperature based on the operational environment and historical data, providing a more tailored and dynamic response to varying conditions.
4. **Multi-Sensor Networks:** Expand the system to support multiple sensors distributed across different parts of a device or facility, enabling comprehensive temperature monitoring and more effective overheating prevention.
5. **Energy Harvesting:** Explore the use of energy harvesting technologies to power the system, particularly in remote or off-grid applications, enhancing its sustainability and reducing dependence on external power sources.

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